Project 2 on Neural networks

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This project ...

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I. INTRODUCTION

II. THEORY

A. Cost/ Loss functions and their gradients

The loss function is the most important function for training this model, since training a model means to minimize the related cost function of the model. To minimize the cost function, we use backpropagation and gradient descent.

1. MSE

with and without L_1 and L_2 norms, Regression problems. L2 norm is for Momentum, L1 is Lasso.

2. Binary cross entropy: log loss

For binary classification problems with and without the L_1 and L_2 norms.

3. Multiclass cross entropy cost function

Softmaxx cross entropy or softmax loss function.

B. Activation functions

1. Sigmoid logit function

2. RELU function

3. leaky RELU function

III. METHODS

IV. RESULTS

V. DISCUSSION

VI. CONCLUSION